Supplementary Report for *"Detecting Root Causes for Process Performance Anomalies Using Causal Inference"*

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I. Evaluation Results of Extended Attributes

Correlation is used as an initial screening criterion for identifying potential causal factors for hypotheses formulation. To assess both correlation and predictive contribution of extended attributes to case duration, we train an XGBoost-based regression model and evaluate feature effects using SHAP (SHapley Additive exPlanations) analysis [1]. SHAP treats each feature as a player in a cooperative game and quantifies its marginal contribution to the model's predictions through Shapley values. Features are then ranked by their mean absolute SHAP values and visualized in a beeswarm plot, which simultaneously conveys each feature's importance and the direction of its influence.

Fig. S1 and Fig. S2 present the SHAP bee-swarm plots for four synthetic event logs and nine real-life event logs. Although the influence of time-related attributes (e.g., month, week, and date), workload-related attributes (e.g., case, event, and resource), and the activity-duration composite attribute varies across event logs, several extended attributes consistently rank among the most influential. This consistency underscores the strong relevance to case duration prediction and supports their inclusion as candidate causal factors in root cause analysis of case duration.

II. Comparison Results for Meta-Learner and Learning Model Combinations on Synthetic Event Logs

Table S1 summarizes the accuracy and execution time of root cause analysis approaches. Accuracy is defined as the proportion of correctly identified root causes relative to the total number of known ground-truth causes. We implemented the proposed approach using four meta-learning algorithms S-Learner, T-Learner, X-Learner, and R-Learner, denoted as TRCA(S), TRCA(T), TRCA(X), and TRCA(R), respectively. Furthermore, we compared the XGBoost and LightGBM as base learners and evaluate both regression and classification formulations, denoted as TRCA(·)-Reg and TRCA(·)-Class, respectively. In TRCA(·)-Class, the anomaly type is case duration timeout and the timeout threshold is 3 days. In contrast, TRCA(S)-Reg predicts the remaining time of a process instance to explore factors influencing case duration. For the comparison baseline, AITIA_PM(SPC) adopts the same 3-day timeout threshold as TRCA(·)-Class, as it is limited to a binary outcome setting.

As shown in Table S1, XGBoost trades some computational efficiency for higher accuracy in root cause identification compared with LightGBM. A possible explanation is that XGBoost exhibits better stability and robustness when handling noisy data. Compared with the T-Learner, X-Learner, and R-Learner, the S-Learner achieves the best efficiency while maintaining accuracy. This advantage arises because it uses a single model for all samples and shares the representation between the treatment and control groups, which typically improves generalization under limited sample sizes or sparse high-dimensional features. Accordingly, we adopt the S-Learner and XGBoost combination for TRCA in subsequent experiments. The baseline approach AITIA_PM(SPC) underperforms this combination in both accuracy and computational efficiency.

III. Visualization of the Causal Graph

Taking the root cause results obtained by TRCA(S)-Class in the SimLog1 log as an example, the corresponding causal graph is shown in Fig. S3. It illustrates both the case-level and the event-level root causes that lead to case timeouts. As shown in Fig. S3, caseLoad, month, and the timeout acitivities "Take in charge ticket" and "Insert ticket" are identified as case-level root causes of process delay. Further tracing back along the causal paths uncovers time-level root causes such as eventLoad, resource, hour, and day. Among these, the intrinsic relationship between resourceLoad and resource suggests that resource load is a primary factor contributing to execution timeouts.

IV. Comparative Experiments and Visual Analysis in Real-life Event Logs

To further demonstrate the effectiveness of the proposed approach, we leverage the root cause analysis results from [2]. The work introduces a root cause analysis approach based on structural equation models (SEM) to investigate case duration timeouts in the CoSeLoG log. However, SEM cannot simultaneously handle discrete categorical attributes and continuous numerical attributes, and its granularity of potential causal factors differs from that of the proposed approach. Therefore, SEM is not incorporated into the TRCA approach. To ensure a fair comparison, the root cause related attributes identified by AITIA_PM(SPC) and TRCA(S) are

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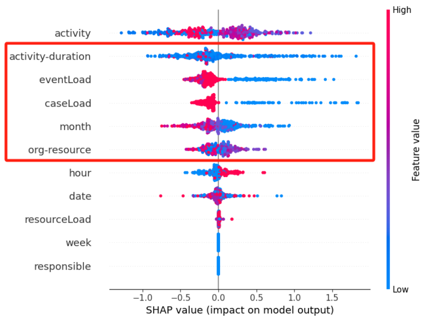
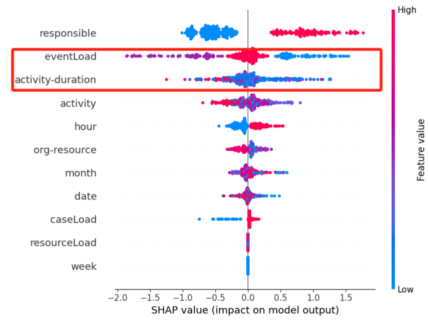
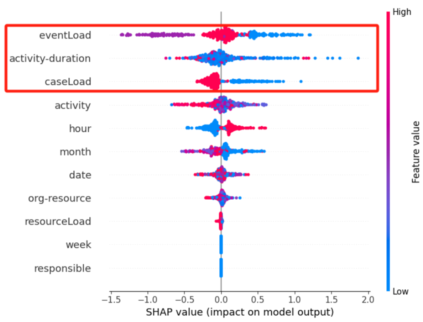


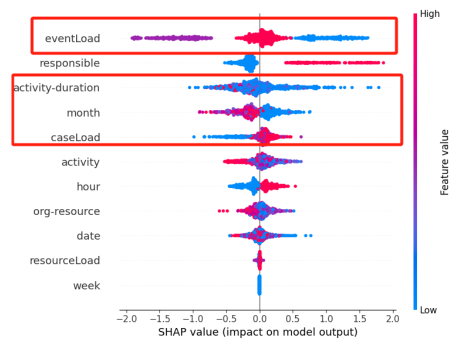
(a) SimLog1



(b) SimLog2



(c) SimLog3



(d) SimLog4

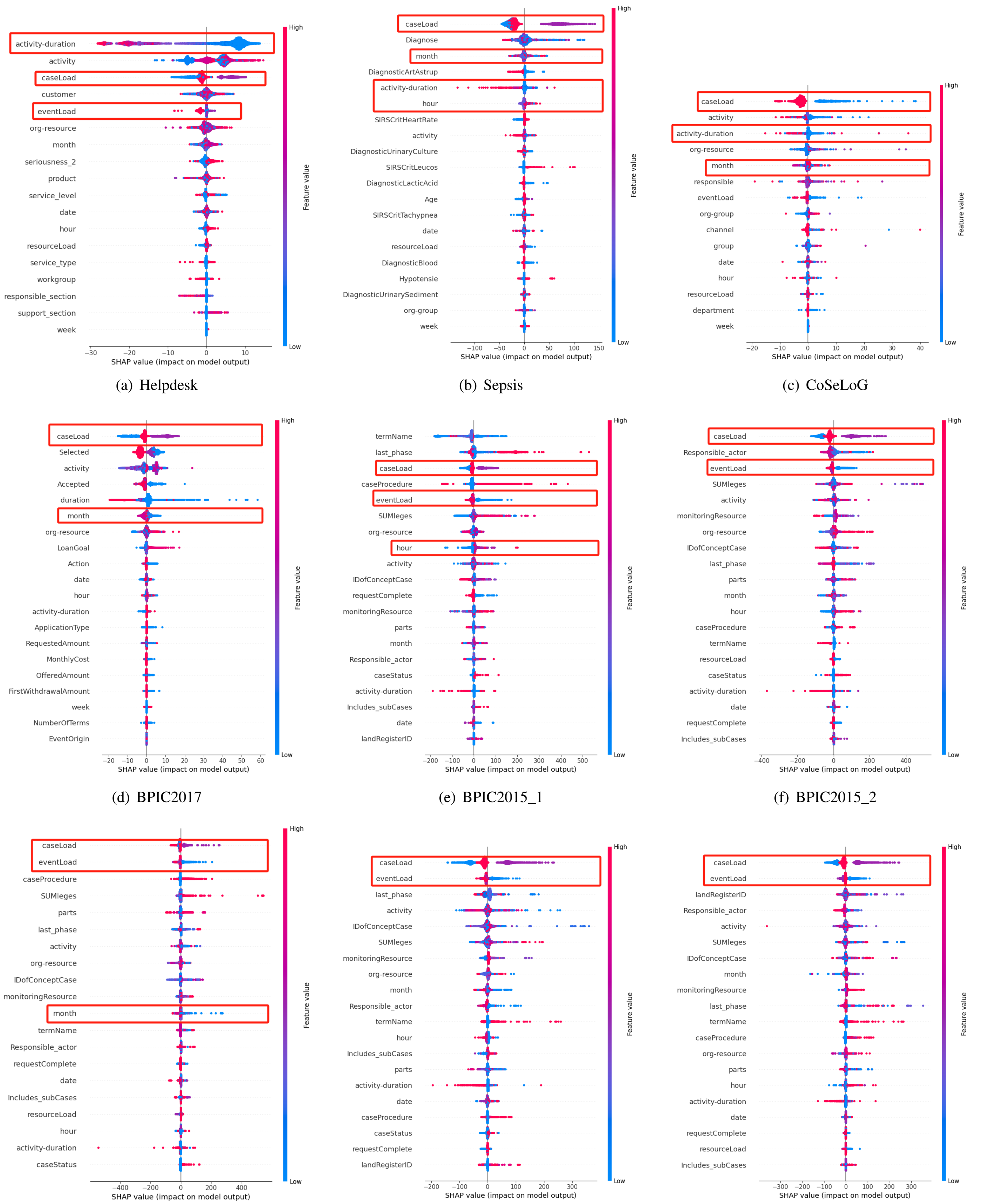
Fig. S1: SHAP Bee-swarm Plots of Feature Importance in Synthetic Event Logs.

TABLE S1: Comparison of accuracy and execution time for case duration anomaly in synthetic event logs

Data set		**SimLog1**		**SimLog2**		**SimLog3**		**SimLog1**	
Known root causes		**Take in charge ticket (res7)**		**responsible (2)**		**caseLoad**		**Take in charge ticket (res7) responsible (2) caseLoad**	
Metrics		Accuracy	Time(s)	Accuracy	Time(s)	Accuracy	Time(s)	Accuracy	Time(s)
XGBoost	TRCA(S)-Class	1	10	1	15	1	7	1	16
	TRCA(S)-Reg	1	4	1	7	1	7	1	7
	TRCA(T)-Class	1	12	1	18	1	14	1	20
	TRCA(T)-Reg	1	6	1	15	1	12	1	13
	TRCA(X)-Class	0	80	1	121	1	99	1	115
	TRCA(X)-Reg	1	57	1	96	1	98	1	103
	TRCA(R)-Class	1	106	1	107	1	87	1	117
	TRCA(R)-Reg	1	100	1	67	1	88	1	92
LightGBM	TRCA(S)-Class	0	7	1	10	1	7	1	13
	TRCA(S)-Reg	0	3	0	3	1	6	0.66	6
	TRCA(T)-Class	1	8	1	13	1	10	1	14
	TRCA(T)-Reg	0	3	0	5	0	7	0.33	7
	TRCA(X)-Class	1	60	1	72	1	56	1	99
	TRCA(X)-Reg	0	41	0	47	0	84	0.33	61
	TRCA(R)-Class	0	113	1	143	1	80	0	134
	TRCA(R)-Reg	1	99	1	66	1	83	1	113
AITIA_PM(SPC)		0	23	1	15	1	8	0.33	48

used as input features for the SEM model. The same timeout threshold for case duration is applied, and the resulting causality diagram is shown in Fig. S4. The diagram reveals that the case-level attributes "department", "responsible", and "Process_workload" exhibit causal relationships with case duration timeout, where the "Process_workload" corresponds to "caseLoad". Although SEM does not determine the causality direction, the consistency of its inference results with those of TRCA(S) supports the comprehensiveness and validity of the root causes identified by the proposed approach for the CoSeLoG log.

For the BPIC2015 series logs, consistent root causes identified by both TRCA(S) and AITIA_PM(SPC) include "caseLoad" and "caseProcedure(Uitgebreid)", indicating that despite originating from different departments, the factors influencing case duration are similar. Taking the BPIC2015_3 log as an example, we further evaluate the rationality of both approaches through visualization. Although the BPIC2015_3 log contains a wide range of activities, variants, and attributes, its relatively small data scale poses challenges for reliable analysis. TRCA(S) identifies two root causes, whereas AITIA_PM(SPC) identifies 17 root causes, with the results



(a) Helpdesk

(b) Sepsis

(c) CoSeLoG

(d) BPIC2017

(e) BPIC2015_1

(f) BPIC2015_2

(g) BPIC2015_3

(h) BPIC2015_4

(i) BPIC2015_5

Fig. S2: SHAP Bee-swarm Plots of Feature Importance in Real-life Event Logs.

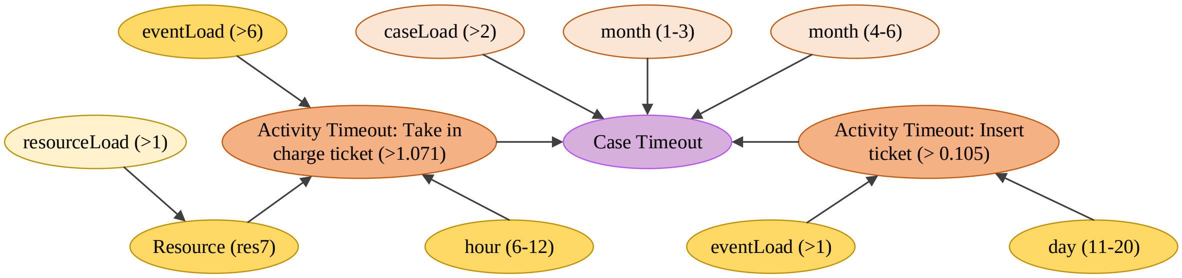
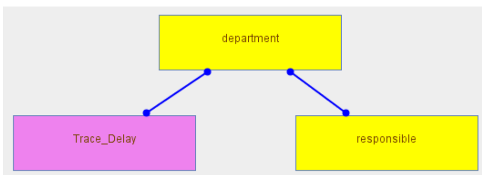
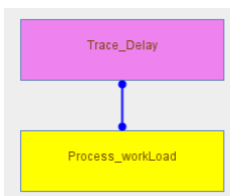


Fig. S3: Causal graph in SimLog1 log using TRCA(S)-Class.

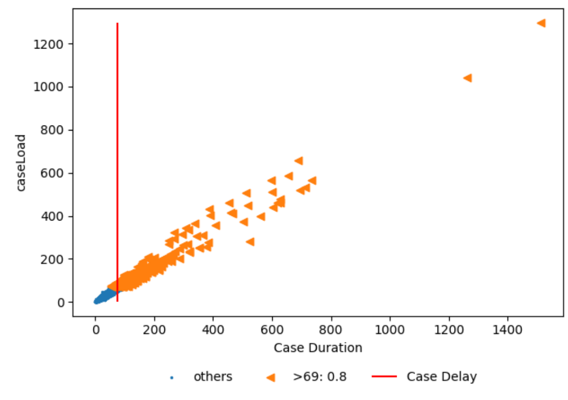


(a) Discrete categorical attributes

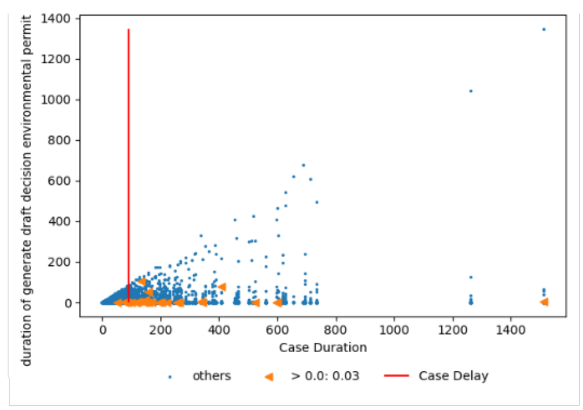


(b) Continuous numerical attribute

Fig. S4: SEM results of case duration timeout anomaly in the CoSeLoG log



(a)



(b)

Fig. S5: Visualization of case duration timeout anomaly in the BPIC2015_3 log

of TRCA(S) included in the latter. One common root cause, “caseLoad”, is visualized in Fig. S5(a). The legend displays the ATE estimated by TRCA(S), demonstrating a positive correlation between “caseLoad” and case duration timeout. However, TRCA(S) relies on learning models, and the accuracy of estimating counterfactual results may be compromised by insufficient training data, potentially resulting in incomplete root cause analyses. A different root cause, the duration timeout of the activity “generate draft decision environmental permit” is visualized in Fig. S5(b), where the legend represents the DFR calculated by AITIA_PM(SPC). The visualization shows that this activity contributes to a higher likelihood of case duration timeout. Despite AITIA_PM(SPC) identifying 15 root causes related to “activity-duration”, only two can be further traced back, which may be attributed to the limited data scale and the absence of complete potential causal factors in the BPIC2015_3 log. Another possible explanation is that AITIA_PM cannot handle confounders well, and the root causes obtained in complex situations are inaccurate.

References

[1] E. Mosca, F. Szigeti, S. Tragianni, D. Gallagher, and G. Groh, “Shap-based explanation methods: a review for nlp interpretability,” in *Proceedings of the 29th international conference on computational linguistics*, 2022, pp. 4593–4603.

[2] M. S. Qafari and W. van der Aalst, “Root cause analysis in process mining using structural equation models,” in *International Conference on Business Process Management*. Springer, 2020, pp. 155–167.